Contents

Supplementary Material S4: UML-to-Ecology Terminology Mapping

Bridging Software Engineering and Ecological Concepts

Introduction

This document provides a comprehensive mapping between UML terminology and ecological modeling concepts. It is designed to help ecologists interpret UML diagrams and software engineers understand ecological context.

Class Diagram Terminology

UML Term	Definition	Ecological Equivalent	Example
Class	A template defining attributes and behaviors	Population compartment, state variable, or entity type	`Larva` class represents larval stage
Object	An instance of a class	Individual organism or cohort	A specific larval cohort on day 15
Attribute	A property of a class	State variable, parameter	`abundance: float` = population size

UML Term	Definition	Ecological Equivalent	Example
Method/Operation	A behavior of a class	Process, rate calculation	`develop(temp)` = development function
Association	Relationship between classes	Ecological interaction	Temperature affects Larva
Aggregation (◇)	"Has-a" relationship (loose)	Contains, hosts	BreedingSite has Eggs
Composition (◆)	"Has-a" relationship (strong)	Integral component	Environment contains Temperature
Inheritance (△)	"Is-a" relationship	Taxonomic or functional grouping	Larva is-a LifeStage
Abstract class	Template that can't be instantiated	Generalized category	`LifeStage` (not instantiated directly)
Interface	Contract specifying required behaviors	Functional role	`Developable` interface
Multiplicity	How many instances relate	Population size constraints	`"many"` larvae in `"1"` breeding site
Visibility (-, +)	Access level (private, public)	Internal vs. observable	- internal rate, + observable abundance

Ecological Interpretations

Class = Compartment

```
┌─────────────┐
│   Larva     │  ← This represents the larval stage compartment
├─────────────┤     in a stage-structured model
│ - N: float  │  ← N(t) = larval abundance at time t
│ - μ: float  │  ← μ = mortality rate parameter
├─────────────┤
│ + die()     │  ← Mortality process
│ + develop() │  ← Development process
└─────────────┘
```

Inheritance = Stage Hierarchy

```
      LifeStage         ← Shared properties of all stages
          △
      ┌───┴─────┐
  Aquatic    Terrestrial   ← Habitat-based grouping
      △          △
    ┌─┴─┐        │
Egg Larva     Adult       ← Specific stages
```

State Machine Terminology

UML Term	Definition	Ecological Equivalent	Example
State	A condition or mode	Life stage, physiological state	`Larva`, `Bloodfed`, `Diapausing`
Transition	Change from one state to another	Development, mortality, behavior change	Larva → Pupa
Guard [condition]	Condition that must be true	Environmental threshold, trigger	`[temp > 10°C]`
Action / effect	What happens during transition	Process execution	`/ accumulateDD()`
Entry action	What happens when entering state	Initialization	`entry / resetDegreeDays()`
Do activity	Ongoing behavior in state	Continuous process	`do / feed()`
Exit action	What happens when leaving state	Finalization	`exit / recordDuration()`
Initial state (●)	Starting point	Birth, introduction	Egg from oviposition
Final state (◉)	Termination point	Death, removal	Mortality
Composite state	State containing substates	Stage with internal structure	Larva with instars L1-L4
Fork/Join	Parallel state activation	Simultaneous processes	Development AND feeding

Ecological Interpretations

State = Life Stage or Physiological Condition

```
┌─────────────────┐
│   Bloodfed      │  ← Physiological state of female adult
│                 │
│ entry/start     │  ← Begin gonotrophic cycle
│   gonotrophic   │
│   Cycle()       │
│                 │
│ do/developEggs  │  ← Ongoing egg development
│   (temp)        │
└─────────────────┘
         │
         │ [cycle complete]  ← Guard: when eggs mature
         ▼
┌─────────────────┐
│     Gravid      │  ← Ready to oviposit
└─────────────────┘
```

Guard = Environmental Threshold

```
Egg ──[temp > 10°C AND moisture > 0.5]──> Larva
         │
         └── This guard specifies:
             - Minimum temperature for hatching
```

```
- Minimum moisture for survival
- Both must be true simultaneously
```

Activity Diagram Terminology

UML Term	Definition	Ecological Equivalent	Example
Activity	A unit of work	Process, calculation	"Calculate mortality"
Action	An atomic step	Single operation	"Update temperature"
Control flow (→)	Sequence of activities	Process ordering	Development before mortality
Decision node (◇)	Branching point	Conditional logic	If gravid, then oviposit
Merge node (◇)	Joining branches	Convergence	All paths lead to output
Fork bar (═)	Start parallel activities	Simultaneous processes	Development ‖ Mortality calc
Join bar (═)	Wait for all parallel	Synchronization	Wait for all rates
Partition/Swimlane	Responsibility grouping	Stage-specific processes	"Larval dynamics" partition
Object node	Data being passed	Variable, parameter	Temperature value
Initial node (●)	Start of workflow	Time step begins	Start of daily update
Final node (◉)	End of workflow	Time step complete	End of daily update

Ecological Interpretations

Activity = Simulation Process

```
┌─────────────────────────┐
│ Calculate development   │  ← This activity computes
│ rates                   │     stage-specific development
│                         │     based on temperature
│ Input: Temperature      │
│ Output: devRate[stage]  │
└─────────────────────────┘
```

Fork/Join = Parallel Processes

```
       ═══════════════════  ← Fork: start parallel
       │         │        │
       ▼         ▼        ▼
    ┌───────┐ ┌───────┐ ┌───────┐
    │ Eggs  │ │Larvae │ │ Pupae │  ← Independent stage updates
    └───┬───┘ └───┬───┘ └───┬───┘
```

```
 │          │         │
 ═══════════════════    ← Join: synchronize
            │
            ▼
     ┌──────────┐
     │  Output  │
     └──────────┘
```

Sequence Diagram Terminology

UML Term	Definition	Ecological Equivalent	Example
Lifeline	Object over time	Component through simulation	Larva population lifeline
Message (→)	Communication between objects	Data/signal passing	Temperature sends rate to Larva
Activation bar	Period of activity	Active processing time	Larva processing development
Return message (⟵)	Response to a call	Result, feedback	Larva returns survival prob
Self-message	Object calls itself	Internal calculation	Larva calculates mortality
Alt fragment	Alternative paths	Conditional behavior	If DD >= threshold then advance
Loop fragment	Repeated execution	Iteration	For each instar
Opt fragment	Optional execution	Conditional process	If gravid, oviposit
Par fragment	Parallel execution	Simultaneous processes	Update all stages

Ecological Interpretations

Lifeline = Component Through Time

```
   Larva          Temperature
     │                  │
     │  getDevRate()    │
     │─────────────────>│  ← Larva requests development rate
     │                  │
     │    devRate       │
     │<─────────────────│  ← Temperature returns rate
     │                  │
     │  develop()       │
     │────┐             │  ← Larva updates internal state
     │<───┘             │
     │                  │
```

Alt Fragment = Conditional Branching

```
    Larva
      │
   ┌──┴──────────────────────┐
   │ alt [DD >= threshold]   │
   ├─────────────────────────┤
   │   advanceInstar()       │  ← If threshold met
   ├─────────────────────────┤
   │ [else]                  │
   │   continue()            │  ← Otherwise continue
   └─────────────────────────┘
```

Common UML-Ecology Mappings

Population Dynamics Concepts

Ecological Concept	UML Representation
Stage-structured population	Class hierarchy with LifeStage parent
Vital rates	Class attributes (developmentRate, mortalityRate)
Density dependence	Method with population density parameter
Environmental forcing	Association with Environment/Temperature class
Carrying capacity	Attribute in habitat/BreedingSite class
Fecundity	Method in Adult class returning egg count
Immigration/Emigration	Methods or transitions to/from external states

Process Representations

Ecological Process	UML Element
Development	Transition with degree-day guard
Mortality	Transition to final state or method call
Reproduction	Message from Adult to Egg (sequence) or oviposit() method
Diapause	Composite state with entry/exit conditions
Competition	Dependency relationship or density parameter
Predation	Association with Predator class

Model Structure

Model Component	UML Element
State variable	Class attribute (e.g., `abundance: float`)
Parameter	Class attribute or constant

Model Component	UML Element
Forcing function	Method in Environment class
Time step	Activity diagram workflow
Feedback loop	Circular message sequence
Spatial unit	Separate class or package

Quick Reference Card

Reading Class Diagrams

```
┌─────────────────┐
│ ClassName       │  → What population/component?
├─────────────────┤
│ - attribute     │  → What state variables?
├─────────────────┤
│ + method()      │  → What processes?
└─────────────────┘

A ───> B          → A affects/uses B
A ◇─── B          → A contains B (loosely)
A ◆─── B          → A contains B (tightly)
A ──△  B          → A is a type of B
```

Reading State Machines

```
● → Start here (birth/introduction)
┌───┐
│   │ → Current state (life stage, condition)
└───┘
──[condition]──> → Transition when condition true
◉ → End here (death/removal)
```

Reading Activity Diagrams

```
● → Begin time step
:Action; → Do this process
◇ → Make decision
═ → Start/end parallel
◉ → End time step
```

Reading Sequence Diagrams

```
│ → Component exists over time
────> → Sends data/signal
<──── → Returns result
┌──┐
│  │ → Active (processing)
└──┘
```

Glossary

Term	Domain	Definition
Aggregation	UML	Whole-part relationship where parts can exist independently
Attribute	UML	Property of a class; maps to state variable or parameter
Cohort	Ecology	Group of individuals of same age/stage
Compartment	Ecology	Subdivision of population (e.g., by stage)
Composition	UML	Strong whole-part relationship; parts cannot exist alone
Degree-day	Ecology	Heat accumulation unit for development
Guard	UML	Boolean condition enabling a transition
Lifeline	UML	Vertical line showing object existence over time
Multiplicity	UML	Number of instances in a relationship
State variable	Ecology	Quantity that changes over time (e.g., abundance)
Stereotype	UML	Extension mechanism (e.g., `<<stochastic>>`)
Transition	UML	Change from one state to another
Vital rate	Ecology	Demographic rate (birth, death, development)

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